

Total No. of Questions : 8]

SEAT No. :

PC2810

[6352]-34

[Total No. of Pages : 2

S.E. (Electronics & Computer Engineering)

COMPUTER ORGANIZATION

(2019 Pattern) (Semester - III) (204203)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Neat diagrams must be drawn wherever necessary.

- Q1)** a) Compare memory mapped I/O with I/O mapped I/O. [5]
b) What is need of DMA? How it works? [6]
c) What are the evolutionary steps of I/O channel? Explain types of I/O channel? [6]

OR

- Q2)** a) Explain data transfer modes in DMA. [5]
b) What is the use of DMA? Explain cycle stealing in DMA? [6]
c) Differentiate between programmed I/O and interrupt driven I/O. [6]

- Q3)** a) What is displacement addressing? Explain its types with calculation of effective address. [5]
b) Explain different addressing modes with example. Discuss drawback of relative addressing mode. [6]
c) Explain following addressing modes of 8086 with suitable examples [6]
i) Index Addressing
ii) Register Indirect
iii) Relative Addressing

OR

- Q4)** a) What is machine instruction? Explain types of instruction. [5]
b) List the features of 8086 microprocessors. [6]
c) Draw and explain the functional block diagram of 8086. [6]

P.T.O.

- Q5)** a) Explain hardware organization and execution of 4 stage instruction pipeline.[5]
b) What are various hazards in instruction pipelining? Explain with example.[6]
c) Explain the instruction cycle in detail. [7]

OR

- Q6)** a) What are the advantages of pipelining? Define latency and throughput of pipeline. [5]
b) What is register organization? What are different types of registers? Explain in detail. [6]
c) Explain the use of the following registers of 8086 CPU: [7]
i) General purpose registers
ii) Segment Register
iii) Flag Register

- Q7)** a) Write control sequence for execution of the instruction Add (R3), R1.[5]
b) What are the different design methods for Hardwired control units? Explain any one. [6]
c) What is microprogrammed control unit design? Draw and explain basic structure of micro-programmed control unit. [7]

OR

- Q8)** a) Compare hardwired control and microprogrammed control unit. [5]
b) Write a control sequence for the following instruction for single bus organization: SUB (R3), R1. [6]
c) Explain in detail the following microinstruction sequencing techniques:[7]
i) Single Address Fields
ii) Variable Address Fields

x

x

x